Fiscal Year:	FY 2005	**Task Last Updated:**	FY 01/24/2005
PI Name:	Andrews, J. Barry Ph.D.		
Project Title:	Coupled Growth in Hypermonotectics		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Basics of solidification		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Birmingham	**State:**	AL
Zip Code:	35294-4461	**Congressional District:**	7
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	OTHER
Start Date:	01/01/1993	**End Date:**	05/31/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	6	**No. of Master' Degrees:**	6
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	McFadden, Geoffrey B. (National Institute of Standards and Technology)		
Grant/Contract No.:	NAG8-99059		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	This project includes both a modeling and experimental segment. A major portion of the project involves the development of models describing events at a monotectic (and hypermonotectic) solidification front. However, the modeling segment also involves the determination of permissible gravity level limits and required thermal gradients for proper processing of samples. An extensive experimental segment is also underway. This segment requires ground-based experimentation and testing, ampoule design and testing, and microgravity processing. Ampoule development has resulted in a design that can control thermal end effects and undesirable wetting tendencies while eliminating the formation of free surfaces during microgravity processing. Several preliminary microgravity experiments have been carried out to date. The objective of this investigation is to gain an improved understanding of solidification processes in immiscible alloy systems. A portion of this study involves experimentation that will permit steady-state coupled growth of hypermonotectic-composition samples to produce aligned microstructures. A parallel effort is underway to develop a model for the coupled growth process in monotectic systems. This analysis started with the basic		

Task Description:	equations for diffusion-controlled growth and avoided many of the simplifying assumptions often utilized in similar analyses. The results obtained from experimentation will be compared to predictions from the model and utilized to improve the model. In order to permit steady-state coupled growth in hypermonotectic-composition samples, experimentation must be carried out under low-gravity conditions. Many alloys in immiscible systems show great promise for potential applications in areas that include superconductors, magnetic materials, catalysts, and electrical contacts. However, there are many details of the solidification process that are poorly understood in immiscible alloys because these details are masked by gravity-driven phenomena. This project is aimed at using the unique environment available in space to improve this understanding in order to make possible the production of new materials using specialized processing techniques. The significance of this project lies in the scientific gains to be made in truly understanding the coupled growth process in immiscible alloys.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Immiscible alloys show promise for use in many applications. For example, some immiscible alloys may be used to produce filters that can remove micron-sized particles for medical applications. Other alloys are expected to exhibit superconducting characteristics or have unusual magnetic properties. Proper development and use of these intriguing alloys could have a significant and positive impact on mankind.
Task Progress:	This past year saw several changes in the Coupled Growth in Hypermonotectics project. During the spring of 2004 the PI was informed that due to NASA's new focus on the Exploration Initiative most materials science investigations would be terminated. The PI was provided funding through the end of the fiscal year at the normal rate and was told the project would be scaled back to a ground based effort for FY 2005. These changes have obviously impacted the project and the work being undertaken. Two of the graduate students supported by this effort left the program this year due to uncertainties in funding. Efforts are currently focusing on modeling and on experimentation that can be carried out on ground. In one segment of the investigation work is underway to identify the cause of formation of the doubly oriented arrays of droplets that are frequently observed during directional solidification of hypermonotectics. One hypothesis proposes that this droplet formation occurs due to breakdown of the liquid fibers in the elevated temperature region of the sample near the solidification front. However, this hypothesis cannot account for the alignment that is observed transverse to the growth direction. It has also been postulated that the doubly oriented structure only occurs when processing conditions approach the interface stability limit. A carefully selected experimental matrix is being used to assess conditions over which this unique microstructure forms. Another effort is focused on modeling and measuring the influence of thermal gradient during directional solidification on interfiber spacings in monotectic composition alloys. It has been predicted that surface tension driven flow at the growing fiber tips will influence the redistribution of solute during the coupled growth process and, as a result, will impact the spacing between fibers. Modifications have been made to a directional solidification furnace that will permit variations in the thermal gradient with little impact on solidification rate. Experimentation is currently underway. Another task involves modeling and experimentally verifying characteristics related to the wetting behavior of immiscible melts. It has been observed that structures produced during low-gravity processing can be influenced by the wetting characteristics between the immiscible phases and the container used during processing. This portion of the investigation is looking at variations in behavior as alloy composition is varied across the miscibility gap and is almost complete The last active task involves modeling and experimentally verifying the flow fields that develop in advance of the solidification interface in a hypermonotectic alloy. A transparent metal analog is being used for this portion of the investigation so that microstructures and flow fields can be observed during directional solidification. This portion of the investigation is also almost complete. A prototype sample/ampoule assembly was tested this past year by processing in the Quench Module Insert breadboard furnace at Marshall Space Flight Center. The data obtained from this processing run permitted testing of thermal models used to estimate furnace performance and provided promising results for meeting science requirements. This experimentation also permitted evaluation of the piston/spring assembly in the prototype ampoules and control of thermal end effects. A new ampoule sealing method was developed and extensively tested that produces a reliable metallic hermetic seal. A facility has been constructed that will permit extended elevated temperature, high-vacuum stability testing of ampoule materials used in the Coupled Growth in Hypermonotectics project. This facility will allow accumulation of any gaseous byproducts over an extended time period and identification of these gases through residual gas analysis. However, it is no longer clear whether this data will be needed.
Bibliography Type:	Description: (Last Updated: 03/14/2008)
Abstracts for Journals and Proceedings	Andrews JB, Liu QQ, McFadden GB. "Monotectic Growth: Unanswered Questions." Fourth International Conference on Solidification and Gravity, Miskolc-Lillafured, Hungary, Sept 6-10, 2004. Abstract Book, Fourth International Conference on Solidification and Gravity, Miskolc-Lillafured, Hungary, Sept 6-10, 2004. page 4. , Sep-2004
Presentation	Andrews, J. B.; Downs, D. A. and Liu, Q. Q. "Monotectic Growth, Unanswered Questions" Fourth International Conference on Solidification and Gravity, Miskolc-Lillafured, Hungary Sep-2004